Yumiao Xiao

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Address: No.1 Du Xue Rd, Nansha District, Guangzhou, China

Research Interests

Terrestrial Carbon Cycle, Urban Ecosystem and Global Change, Grassland Ecology, Remote Sensing, GIS

Education

The Hong Kong University of Science and Technology (Guangzhou), China 2024/09 – 2026/07 (Expected)

- **Master of Philosophy** in Urban Governance and Design
- Advisor: Dr. Wufan Zhao
- Thesis: Precise Carbon Accounting from 3D Urban Expansion

Beijing Normal University, China 2020/09 – 2024/07

- **Bachelor of Science** in Geography
- Advisor: Dr. Yang Yang, Dr. Bingwei Zhang
- Thesis: Response Patterns and Mechanisms of Grassland Ecosystem Productivity to Changing Precipitation

Research Experience

Research Assistant, Beijing Normal University, China 2022/10 – 2024/07

- Advanced Institute of Natural Sciences
- Advisor: Dr. Yang Yang
- Studying fractional vegetation coverage estimation and its spatiotemporal patterns and causal mechanisms

Research Assistant, Beijing Normal University at Zhuhai, China 2022/05 – 2024/07

- Zhuhai Branch of State Key Laboratory of Earth Surface Processes and Disaster Risk Reduction
- Advisor: Dr. Bingwei Zhang
- Investigating grassland productivity responses to drought using field observations and manipulative experiments

Publications

[1] **Xiao, Y.**, Jiang, Y., Li, J., Lu, H., Wu, X., Jiang, Y., & Zhang, B. (2025). Aridity-dependent response bias of grassland productivity to precipitation in experiments compared to observations. *Ecology*, 106(9), e70189.

[2] **Xiao, Y.**, Li, H., Li, H., Liu, S., Hu, W., Xiao, C., & Xie, Y. (2024). GIS field practice design for land use mapping: A case study in Xiafu Village, Shaoguan City. *Education of Geography*, 08: 70-73. (In Chinese)

[3] **Xiao, Y.**, Lv, W., & Zhao, W. (2025). Exploring multiscale variations in greenspace exposure drivers: A perspective on the modifiable areal unit problem. *IGARSS 2025 - 2025 IEEE International Geoscience and Remote Sensing Symposium*. (In Press)

[4] Patent: A device for automatic measurement of plant response curves to the environment. (No.ZL202220338450.1)

[5] **Xiao, Y.**, Zhang, B., Cadotte, M. W., Wu, X., Jiang, Y., Chen, S., & Chu, C. Plant size rather than density regulates grassland aboveground productivity responses to drought. *Global Ecology and Biogeography*. (Under Review)

[6] Lv, W., Dai, S., Song, Y., Zhao, W., Yi, W., **Xiao, Y.**, & Jia, N. Measuring causal strengths by geographical cross mapping cardinality. *International Journal of Geographical Information Science*. (Revised & Resubmitted)

[7] **Xiao, Y.**, Yang, Y., Zhao, T., Wang, W., Lv, W., & Zhao, W. Township-scale vegetation greening in the Guangdong-Hong Kong-Macao Greater Bay Area: Pathways, spatiotemporal heterogeneity, causality, and driving mechanisms. *International Journal of Applied Earth Observation and Geoinformation* (Manuscript to be Submitted)

[8] Ren, Y., Luo, J., **Xiao, Y.**, Yang, G., Fu, Y., Zhao, W., Chen, Y., Chen, Y., & Qin, J. Discovering the changes in spatial fairness of global urban greenery. *Sustainable Cities and Society* (Manuscript to be Submitted)

[9] **Xiao, Y.**, Wang, T., Liu, J., Lv, W., Dai, S., & Zhao, W. Indirect environmental effects of urbanization mitigate carbon-warming feedback in China. (In Preparation)

[10] **Xiao, Y.**, Wang, T., Li, X., Dai, S., & Zhao, W. Quantifying urban area- and height-induced carbon emissions by ensemble learning experiments. (In Preparation)

Conference Presentation

Xiao, Y. et al., Exploring multiscale variations in greenspace exposure drivers: A perspective on the modifiable areal unit problem. IEEE International Geoscience and Remote Sensing Symposium (IGARSS), Brisbane, Australia. August 2025 (Oral Presentation)

Honors & Awards

Research Travel Grant (RMB10,000), The Hong Kong University of Science and Technology (Guangzhou) 2025
Red Bird MPhil Postgraduate Scholarship (RMB10,000/month), The Hong Kong University of Science and Technology (Guangzhou) 2024-2026
Academic Excellence Scholarship (RMB2,000/year), Beijing Normal University 2022-2023
Outstanding Undergraduate Research Training Program (RMB2,000), Beijing Normal University at Zhuhai 2022
Outstanding Freshman Scholarship (RMB10,000), Huitong College, Beijing Normal University 2020

Skills

Language: Mandarin Chinese (Native), English (Fluent)
Computer: R, Google Earth Engine, ArcGIS, QGIS, GeoDa

Last updated: 2025/11/21